

Linus Dominic Nathaniel

Lead Data Engineer / Data Architect

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Open to EU relocation (visa sponsorship required) and to EU-remote roles

SUMMARY

Data engineer and architect with 6+ years building platforms on top of messy operational systems. I currently lead a team of 4–6 engineers at Altair Capital Group, a US real-estate finance firm, where I own the data architecture, the AWS footprint the whole platform runs on, and every third-party data integration we depend on. Most of my work comes down to reconciliation: getting several systems that each think they own the customer to agree on one version of them. Comfortable across Redshift, BigQuery and Athena over S3 parquet, Airflow, dbt, Spark and Glue, and equally at home in the application layer when the data problem turns out to be a product problem. Before Altair I worked in-house at two product companies, including a multi-vendor marketplace platform.

LEADERSHIP

Team Lead, Altair Capital Group (Dec 2025 – present). 4–6 engineers across data platform, BI, application and integrations.

- I run architecture and code review, plan the sprints, and own the standards and PR conventions the team builds to.
- Every initiative listed under Altair below was delivered by this team with me leading the design and the build. Earlier leadership: the Talend to Airflow migration at VentureDive, and Lead Support Engineer at Bleed AI.
- Mentoring engineers into owning their own domains. Involved in hiring and technical screening, and I sit between lending and operations stakeholders and the engineers gathering requirements and negotiating what each system promises to deliver.

EXPERIENCE

Lead Data Engineer / Data Architect, Altair Capital Group

December 2025 – Present | Remote, US real-estate finance firm | Leading 4–6 engineers

Data platform and architecture

- Built the company's unified customer-journey model, which reconciles a parquet data lake, dialer logs and the CRM into one phone-keyed six-stage funnel across roughly 2M records. It is now the source everything downstream reads from.
- Rewrote the S3 data lake layout: canonical zones, sane partitioning, proper cataloging across about 29 ingestion and transformation jobs. Cleared out years of duplicated data, and the zoning convention is still what the team follows.
- Built the reconciliation framework between the loan-origination platform and the warehouse, so operations can see exactly where the two disagree. Diffs are auditable across millions of records; before this the gaps were invisible.
- Moved BI onto Apache Superset and made it deployable from code, using Python and Jinja-templated MySQL views. It powers the Performance Score Ratings, Loan Officer pipeline and Daily Call dashboards that sales, operations and leadership use daily.
- Architected the conversation-intelligence layer behind the company's AI telemarketing agent: an LLM detector for meaningful conversations, plus a call-note generator that writes a different note for the Loan Officer, the Processor and the CCO from the same call, straight back into CRM Leads and Contacts.

Platform engineering and AWS infrastructure

- Built and run the borrower portal and internal loan platform: a Node and PostgreSQL backend with a React 19 and TypeScript front end migrated in as a strangler around the original client. It covers borrower onboarding, loans, document and task workflows, e-signature with generated PDFs, underwriting and processing pipelines, role-based permissions with field-level masking, audit logging and realtime sync over WebSockets.
- Own the whole AWS footprint: EC2 behind nginx, RDS PostgreSQL, S3 for documents and the data lake, CloudFront, Lambda and API Gateway for vendor callbacks, Secrets Manager and SSM for credentials, CloudWatch and SNS for alerting, and the IAM roles and policies behind all of it.
- Moved the platform off a JSON file store onto PostgreSQL as the system of record, across roughly 29 numbered migrations, and wrote the deploy tooling: guarded publish-to-EC2 scripts, boot migrations, separate readiness and liveness endpoints, drain-aware restarts and S3 restore.
- Ran the performance program off measured evidence rather than guesswork: nginx timing logs, pg_stat_statements, RDS Performance Insights, event-loop lag histograms and per-endpoint p50/p95 telemetry. Cut the main hydration payload from 7.75MB to 124KB by replacing one global version counter with per-domain version vectors and partial state.
- Replaced a sequential scan on a 267k-row audit table with the right expression index, taking the activity query from a parallel seq scan to a 0.06ms index scan.
- Built global search on PostgreSQL itself (pg_trgm, tsvector, phone-digit matching) rather than adding a search engine, with encrypted and SSN-typed fields kept out of the index and permissions re-checked against live data on every hit.

Third-party data integrations

- Zoho first: CRM deal-to-loan field and picklist mapping into the portal, plus OAuth-based transactional mail. Then voice and telephony on Twilio and Telnyx (dialer, recordings, browser softphone), and Square for bookings and payments.

- Then the full Cotality (formerly CoreLogic) suite: Instant Merge credit reports, SSA-89 SSN verification, LoanSafe fraud checks, and ValueLink appraisal ordering with 1004, 1025 and 1014 form field mappings against UAD 3.6.
- Built the SSA-89 production callback edge as its own Lambda behind API Gateway: it validates the vendor token and source CIDR, archives every payload to S3 with server-side encryption, relays to the portal over short-lived HMAC headers, and lets the portal do an idempotent order update before returning the vendor ACK. UAT and production stayed separate, and auto-submit stayed off until vendor certification passed.
- Set the house rule for vendor failures: a visible failure beats a silent one. Irreversible paid actions like appraisal orders and card payments are never auto-retried; they land in an ambiguous state a human must reconcile.
- Shipped a scoped public API with per-module API keys, action-level scopes, IP allowlists, rate limits, rotation with grace windows and separate gating for PII reads.
- Owned the internal loan-pricing platform for the Bridge and DSCR products, from requirements with the lending team through to term-sheet output, including the calculation engine and the CRM integration.

Senior Data Engineer, VentureDive

June 2025 – December 2025 | Karachi, Pakistan

- Ran the migration of the team's ETL platform off Talend and onto Airflow, with the standards that came with it: scheduling conventions, observability, DAGs versioned in Git. Redesigning for parallelism brought average runtime down and, more usefully, cut the paging.
- Made Airflow plus dbt plus warehouse the team's reference architecture, with dbt models, tests, sources and lineage adopted across downstream transformations.
- Wrote the multi-cloud deployment standard on Terraform and GitHub Actions, used across AWS, GCP and Azure engagements (S3/Redshift, BigQuery/Dataflow, Synapse/Data Factory). Mentored the junior engineers and turned our Airflow patterns and DAG-testing into written team practice.

Data Engineer, Pure Harvest Smart Farms (product-based agri-tech company)

June 2023 – May 2025 | Remote

- Wrote the ETL behind the company's own product platform in PySpark on AWS Glue. Processing time came down around 30%.
- Automated ingestion from several sources into one AWS data lake, and built out a Medallion lakehouse (bronze, silver, gold) on Redshift: modeling, governance and a fair amount of performance tuning.
- Built the Superset dashboards used across cultivation and supply chain.

Data and Machine Learning Engineer, Arpatech (Pvt) Ltd (product company, marketplace platform)

January 2022 – June 2023 | Karachi, Pakistan

- Built the Custom Data Insights platform for Connection Cloud Marketplace, a multi-vendor cloud-services marketplace. It ingested vendor and buyer transaction data and gave both sides of the marketplace spend analytics and side-by-side comparison of offerings.
- Built an Elasticsearch and Kibana app for the sales team. Retrieval got roughly 50% faster. Also shipped a Smart Mirror IoT product doing facial-recognition attendance, which took the manual error out of tracking.

Lead Support Engineer and Data Analyst, Bleed AI

December 2019 – November 2021 | Karachi, Pakistan

- Ran the data-analysis side of the computer-vision projects, fed findings back into model tuning for better image recognition, and cleaned up the data pipelines for speed without giving up integrity.

SKILLS

- **Architecture and governance:** dimensional / Kimball modeling, data contracts, lineage, MDM patterns, Medallion architecture, data domains, conformed dimensions. **Languages:** Python, SQL, JavaScript / TypeScript, Bash
- **Orchestration:** Airflow, dbt, Prefect, AWS Glue, Step Functions. **Processing:** Spark, PySpark, Pandas, Kafka, Hadoop
- **Warehouse and lakehouse:** Redshift, BigQuery, Snowflake, Athena, Databricks. **Databases:** PostgreSQL (migrations, query plans, pg_trgm / tsvector search, Performance Insights), MySQL, MongoDB, Elasticsearch
- **AWS:** EC2, S3, RDS, Lambda, API Gateway, Glue, Athena, Redshift, Step Functions, EventBridge, CloudFront, CloudWatch, SNS, Secrets Manager, SSM, IAM, ECS, Fargate. **GCP:** BigQuery, Dataflow, GCS. **Azure:** Data Factory, Synapse
- **Infrastructure:** Terraform, Docker, nginx, GitHub Actions, CI/CD. **Application:** Node.js, React 19, TypeScript, REST API design, WebSockets, OAuth, HMAC-signed webhooks, PDF generation
- **Integrations:** Cotality / CoreLogic (Instant Merge, SSA-89, LoanSafe, ValueLink), Zoho CRM and Mail, Twilio, Telnyx, Square, MISMO. **BI:** Superset, Power BI, Grafana, Kibana, Streamlit. **LLM work:** conversation intelligence, note generation, prompt engineering, role-aware agent design

EDUCATION, CERTIFICATIONS AND LANGUAGES

- **BSc Computer Science**, Karachi Institute of Economics and Technology (August 2018 – August 2022). English, fluent. Urdu, native
- Certified Data Engineer and Certified Data Analyst, Karachi AI. Computer Vision and Image Processing, Bleed AI